

Extend correlation DAG algorithm to polynomial time on triangle-free extensive-form games

ChatGPT (gpt-5.6-sol)*

Solution generated on 2026-08-24

Abstract

This paper gives a complete affirmative solution to the question whether the correlation-DAG method can be extended so that optimal NFCCE, EFCCE, and EFCE can be computed in polynomial time in every finite two-player, perfect-recall, triangle-free extensive-form game. The result permits arbitrary chance structure and does not require the game to be timeable.

The proof uses the constructive scaled-extension decomposition of the von Stengel–Forges polytope for triangle-free games. We prove directly from that decomposition that all its vertices are deterministic correlation plans, and hence that the polytope is exactly the correlation-plan polytope. We then give exact trigger-deviation inequalities, dualize their polynomial-size sequence-form continuation problems, and compile the scaled extensions into a valid generalized correlation DAG. The resulting DAG has polynomial size (and, conservatively, $O(N^3)$ arcs for an explicit game tree of size N). For every rational linear terminal objective, the resulting rational linear program has polynomial encoding length and computes an optimal NFCCE, EFCCE, or EFCE. This is a complete solution; no case within the stated hypotheses remains unresolved.

1 The open problem

This paper addresses an open problem posed by Brian Zhang, Gabriele Farina, Andrea Celli, Tüomas Sandholm in *Optimal Correlated Equilibria in General-Sum Extensive-Form Games: Fixed-Parameter Algorithms, Hardness, and Two-Sided Column-Generation* [1] (Economics and Computation (EC), 2022), Section 4.4 (Discussion: Relationship to Triangle-Freeness), page 19; also listed as future research direction 1 in Section 7 (Conclusions and Future Research), page 29.. The website catalogue entry is problem 18 (W4221157708_p0) of the [Open Problems in Operations Research](#) collection.

Determine whether the correlation DAG algorithm can be extended so that, on every triangle-free two-player extensive-form game, it runs in time polynomial in the size of the input game (equivalently, yields a correlation DAG/linear program formulation whose size is polynomial in the input game size) for computing an optimal correlated equilibrium (with respect to a given linear objective).

*Generated by the `base_solver_ni_two_iterations` pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve → A.8 verify → A.9 write-up; run `run_20260824_061342_e641f87f, ec-2it-ultra`); two-iteration run, outcome from iteration 1. Independent verifier assessment: APPROVED with progress score 3/3 (full solution). Maintained by the AI in Mathematics & Operations Research project.

The remainder of this document is the solution produced by the model, reproduced verbatim from the pipeline output.

2 Introduction and statement of the result

The chronological correlation-DAG construction may enumerate exponentially many complete prescriptions even in some triangle-free games. That fact does not preclude a different correlation DAG based on a nonchronological, adaptive fill order. We construct such a DAG.

Throughout, the input is a finite extensive-form game with two strategic players, numbered 1 and 2, and arbitrary chance nodes. Both players have perfect recall. Chance probabilities, utilities, and the terminal objective are rational. Let N denote the combinatorial size of the explicit game tree, including histories, information-set incidences, and actions, and let B denote the total bit length of its rational data.

Two information sets of different players are called connected if some history in one is comparable by ancestry to some history in the other. The triangle-free hypothesis used in the target question is the following: there do not exist distinct $I, I' \in \mathcal{I}_1$ and distinct $J, J' \in \mathcal{I}_2$ such that

$$I \bowtie J, \quad I' \bowtie J, \quad I \bowtie J'. \quad (\text{TF})$$

No restriction on the parent sequences of these four information sets is imposed in (TF).

Theorem 1 (Main theorem). *Let G be any finite two-player perfect-recall extensive-form game satisfying (TF). The chance structure may be arbitrary, and G need not be timeable. Then the following statements hold.*

1. *The correlation-plan polytope of G is the projection of the flow polytope of a generalized correlation DAG having $O(N^2)$ nodes and $O(N^3)$ arcs.*

2. *For each*

$$c \in \{\text{NFCCE}, \text{EFCCE}, \text{EFCE}\},$$

the set of correlation plans satisfying the incentive constraints for concept c has an exact polynomial-size rational linear formulation.

3. *For every rational function g on terminal histories, an equilibrium of concept c maximizing the expected value of g can be computed in time polynomial in $N + B$.*

The same conclusions hold under the weaker local triangle condition of Farina and Sandholm stated in Theorem 2.

The theorem is a complete solution rather than partial progress. It does not assert that the unchanged chronological Team-Belief DAG is polynomial on the class. Instead, it extends the algorithm by using the scaled-extension fill order on triangle-free instances and retaining the original constructor outside that class.

3 Sequences, relevance, and correlation plans

Let Z be the set of terminal histories. For a terminal history z , let $p_c(z)$ denote the product of the probabilities of all chance actions on the path to z . Zero-probability chance actions are allowed.

For player i , let $\mathcal{I}_i$ be the collection of information sets and let A_I be the action set at I . Perfect recall implies that every information set $I \in \mathcal{I}_i$ has a well-defined preceding player- i sequence, denoted by $\sigma(I)$. Define

$$\Sigma_i = \{\emptyset\} \cup \{(I, a) : I \in \mathcal{I}_i, a \in A_I\}. \quad (2.1)$$

The sequence (I, a) records the last information set and action in a player's sequence; its earlier actions are determined recursively by $\sigma(I)$.

If two nodes of a rooted tree have a common descendant, then both lie on the unique root-to-descendant path and are therefore comparable by ancestry. Conversely, two comparable nodes have a common descendant. Thus the common-descendant and ancestry formulations of connectivity agree.

For a nonempty sequence (I, a) , its underlying information set is I . A pair $(\sigma_1, \sigma_2) \in \Sigma_1 \times \Sigma_2$ is *relevant* if at least one sequence is empty or the two underlying information sets are connected. Write

$$R = \Sigma_1 \bowtie \Sigma_2$$

for the set of relevant pairs. We also write $I \bowtie \sigma_2$ when $\sigma_2 = \emptyset$ or I is connected to the underlying information set of σ_2 , and use the symmetric notation $\sigma_1 \bowtie J$.

Relevance is closed under taking sequence predecessors. Indeed, the predecessor information sets supplied by perfect recall occur earlier on the same relevant root-to-node paths.

Since

$$|\Sigma_i| = 1 + \sum_{I \in \mathcal{I}_i} |A_I| = O(N),$$

the number

$$M := |R|$$

satisfies $M = O(N^2)$.

3.1 The von Stengel–Forges polytope

Define

$$V = \left\{ v \in \mathbb{R}_{\geq 0}^R : \begin{array}{l} v[\emptyset, \emptyset] = 1, \\ \sum_{a \in A_I} v[(I, a), \sigma_2] = v[\sigma(I), \sigma_2], \quad \forall I \in \mathcal{I}_1, I \bowtie \sigma_2, \\ \sum_{b \in A_J} v[\sigma_1, (J, b)] = v[\sigma_1, \sigma(J)], \quad \forall J \in \mathcal{I}_2, \sigma_1 \bowtie J \end{array} \right\}. \quad (2.2)$$

This is the von Stengel–Forges polytope. It has $O(N^2)$ variables and polynomially many explicitly writable constraints.

For every $v \in V$ and every relevant pair,

$$0 \leq v[\sigma_1, \sigma_2] \leq 1. \quad (2.3)$$

To see this, nonnegativity and each conservation equality imply that every child coordinate is at most its parent coordinate. Repeatedly replacing either sequence by its predecessor, which is permitted because relevance is predecessor-closed, bounds the coordinate by $v[\emptyset, \emptyset] = 1$.

Let Π_i be the set of pure reduced-normal-form plans of player i . Whenever necessary, a reduced plan is completed arbitrarily at information sets having zero own reach. For a distribution μ on $\Pi_1 \times \Pi_2$, define

$$\xi^\mu[\sigma_1, \sigma_2] = \Pr_{\pi \sim \mu} [\pi_1 \text{ realizes } \sigma_1 \text{ and } \pi_2 \text{ realizes } \sigma_2]. \quad (2.4)$$

Let

$$\Xi = \{\xi^\mu : \mu \in \Delta(\Pi_1 \times \Pi_2)\}. \quad (2.5)$$

For each fixed event that the parent sequences and the other player's sequence are realized, the events corresponding to the actions at an information set partition that event. Marginalization therefore gives all equalities in (2.2), and hence

$$\Xi \subseteq V. \quad (2.6)$$

4 The triangle-free scaled-extension decomposition

For a polytope X , a nonempty polytope Y , and an affine function $h : X \rightarrow \mathbb{R}_{\geq 0}$, define the scaled extension

$$X \triangleleft_h Y = \{(x, h(x)y) : x \in X, y \in Y\}. \quad (3.1)$$

Let

$$\Delta_k = \{y \in \mathbb{R}_{\geq 0}^k : \mathbf{1}^\top y = 1\}.$$

The external structural result required below is stated in precisely the form in which it is used.

Theorem 2 (Farina–Sandholm scaled-extension theorem). *Consider a finite two-player extensive-form game in which both players have perfect recall. Suppose that, for every pair of parent sequences (α, β) , there do not exist distinct $I, I' \in \mathcal{I}_1$ and distinct $J, J' \in \mathcal{I}_2$ satisfying*

$$\sigma(I) = \sigma(I') = \alpha, \quad \sigma(J) = \sigma(J') = \beta,$$

and

$$I \bowtie J, \quad I' \bowtie J, \quad I \bowtie J'. \quad (3.2)$$

Then, up to a permutation of coordinates, the polytope V has an iterated scaled-extension representation

$$P_0 = \{1\}, \quad P_t = P_{t-1} \triangleleft_{h_t} X_t \quad (1 \leq t \leq T), \quad P_T = V, \quad (3.3)$$

with the following properties.

1. Every X_t is either a probability simplex or the singleton $\{1\}$.
2. Every scale is nonnegative on P_{t-1} and has the form

$$h_t(x) = x_q \quad \text{or} \quad h_t(x) = \sum_{q \in P_t^{\text{in}}} x_q, \quad (3.4)$$

where all coordinates on the right-hand side were introduced before stage t .

3. *Nontrivial simplex splits are scaled by a single existing coordinate. Multi-coordinate sums implement deterministic marginal or backfill operations.*
4. *Every relevant coordinate is filled exactly once. The number of blocks and the total number of introduced coordinates are $O(|R|)$, and the decomposition can be constructed in time linear in the number of coordinates once the relevance relation is available.*

The construction includes relevance-filtered star components, isolated rank-zero information sets, a maintained set of already-filled coordinates, and the descendant-backfill operations needed when several information sets of one player occur successively.

This is the constructive decomposition proved in Farina and Sandholm [3, Theorems 2 and 5]. The final sentence records parts of the published construction that are necessary for the stated “filled exactly once” conclusion; it is not an additional hypothesis.

We now check every hypothesis of Theorem 2. The game is finite, has exactly two strategic players, and both have perfect recall by assumption. Condition (TF) prohibits (3.2) even without the equal-parent restrictions, so it implies (3.2). The theorem places no restriction on chance probabilities and makes no timeability assumption. It therefore applies to the present game.

For completeness, the star structure behind the condition can be seen directly. Fix parent sequences (α, β) , and form a bipartite graph with the information sets $I \in \mathcal{I}_1$ satisfying $\sigma(I) = \alpha$ on one side and the information sets $J \in \mathcal{I}_2$ satisfying $\sigma(J) = \beta$ on the other. Put an edge between I and J exactly when $I \bowtie J$. If an edge IJ had both endpoints of degree at least two, one could choose another neighbor I' of J and another neighbor J' of I , producing the three forbidden edges in (3.2). Thus every connected component is a star, a single edge, or an isolated vertex.

At a filled parent pair (α, β) , a player-1 star center I generates a simplex split of the form

$$(v[(I, a), \beta])_{a \in A_I} \in v[\alpha, \beta] \Delta_{|A_I|}. \quad (3.5)$$

If J is an unselected neighbor whose joint descendants have been filled, the missing marginal is backfilled by

$$v[\alpha, (J, b)] = \sum_{a \in A_I} v[(I, a), (J, b)]. \quad (3.6)$$

The omitted conservation equality is then automatic:

$$\begin{aligned} \sum_{b \in A_J} v[\alpha, (J, b)] &= \sum_{a \in A_I} \sum_{b \in A_J} v[(I, a), (J, b)] \\ &= \sum_{a \in A_I} v[(I, a), \beta] = v[\alpha, \beta]. \end{aligned} \quad (3.7)$$

The exact construction in Theorem 2 performs these operations recursively in sequence depth and backfills all affected descendants.

Remark (Why the exact backfill theorem is needed). A one-level immediate-neighbor recursion is insufficient. For example, consider a root information set I of player 1, followed by an information set J of player 2, and, after one action at J , another information set K of player 2. Independently splitting the immediate marginals can fail to enforce

$$v[\emptyset, (K, c)] = \sum_{a \in A_I} v[(I, a), (K, c)].$$

The filled-coordinate set and descendant backfill in Theorem 2 enforce this equality. The proof below invokes the full constructive theorem, not the inadequate one-level recursion.

5 Integrality and exactness of the correlation-plan polytope

We next prove, rather than merely assume, that the scaled-extension polytope is exactly Ξ .

Lemma 1 (Vertices of a scaled extension). *Let X and Y be nonempty polytopes and let $h : X \rightarrow \mathbb{R}_{\geq 0}$ be affine. Then*

$$X \triangleleft_h Y = \text{conv} \{(x, h(x)y) : x \in \text{vert}(X), y \in \text{vert}(Y)\}. \quad (4.1)$$

Consequently, every vertex of $X \triangleleft_h Y$ is one of the displayed generating points.

Proof. First, $X \triangleleft_h Y$ is convex. Indeed, take $(x, h(x)y)$ and $(x', h(x')y')$ in the scaled extension and $\theta \in [0, 1]$. Put

$$\bar{x} = \theta x + (1 - \theta)x', \quad \bar{h} = h(\bar{x}) = \theta h(x) + (1 - \theta)h(x').$$

If $\bar{h} > 0$, then

$$\bar{y} = \frac{\theta h(x)y + (1 - \theta)h(x')y'}{\bar{h}} \in Y,$$

because the two nonnegative coefficients in this convex combination sum to one. If $\bar{h} = 0$, nonnegativity of $h(x)$ and $h(x')$ makes the second coordinate zero, and any $\bar{y} \in Y$ may be used.

Now write

$$x = \sum_r \lambda_r x_r, \quad y = \sum_s \mu_s y_s,$$

where $x_r \in \text{vert}(X)$, $y_s \in \text{vert}(Y)$, and the coefficients are convex weights. Affineness of h gives

$$(x, h(x)y) = \sum_{r,s} \lambda_r \mu_s (x_r, h(x_r)y_s). \quad (4.2)$$

Thus every point of the scaled extension lies in the convex hull in (4.1). The reverse inclusion follows from convexity and the fact that every displayed generator belongs to the scaled extension. $\square$

Proposition 1. *Every vertex of V is integral and hence, by (2.3), is a 0-1 vector.*

Proof. Apply Lemma 1 inductively to (3.3). The only vertex of P_0 is the integral point 1. Suppose all vertices of P_{t-1} are integral. Every vertex of P_t is among the points

$$(x, h_t(x)y), \quad x \in \text{vert}(P_{t-1}), y \in \text{vert}(X_t). \quad (4.3)$$

If X_t is a simplex, then y is a unit vector. If $X_t = \{1\}$, then $y = 1$. In either case, (3.4) shows that $h_t(x)$ is an integer, because it is a coordinate or an integer-coefficient sum of coordinates of the integral vector x . Hence every coordinate in (4.3) is integral. Induction proves that every vertex of $P_T = V$ is integral. Bound (2.3) then makes every coordinate zero or one. $\square$

For player i , define the marginal of $v \in V$ by

$$q_1[\sigma_1] = v[\sigma_1, \emptyset], \quad q_2[\sigma_2] = v[\emptyset, \sigma_2]. \quad (4.4)$$

Lemma 2 (Factorization of binary points). *Let $v \in V$ be a 0-1 point. Then each q_i in (4.4) is the realization plan of a pure reduced-normal-form plan, and for every relevant pair,*

$$v[\sigma_1, \sigma_2] = q_1[\sigma_1]q_2[\sigma_2]. \quad (4.5)$$

Proof. The marginal conservation equations give

$$\sum_{a \in A_I} q_1[(I, a)] = q_1[\sigma(I)] \quad \text{and} \quad \sum_{b \in A_J} q_2[(J, b)] = q_2[\sigma(J)]. \quad (4.6)$$

Because all coordinates are binary, a parent value of one has exactly one child action of value one, while a parent value of zero has no child action of value one. Thus q_i specifies one action at every information set having positive own reach. Choosing arbitrary actions at zero-reach information sets completes it to a pure plan π_i .

It remains to prove (4.5). We use induction on the sum of the two sequence depths. The statement is immediate if one sequence is empty.

Suppose first that $q_1[\sigma_1] = 0$. Repeated conservation along the second sequence gives

$$0 \leq v[\sigma_1, \sigma_2] \leq v[\sigma_1, \emptyset] = q_1[\sigma_1] = 0. \quad (4.7)$$

The required conservation constraints exist because relevance is closed under predecessors. The same argument applies if $q_2[\sigma_2] = 0$.

It remains to consider

$$q_1[\sigma_1] = q_2[\sigma_2] = 1.$$

Write

$$\sigma_1 = (I, a), \quad \sigma_2 = (J, b).$$

Their parent marginals also equal one. By the induction hypothesis,

$$v[\sigma(I), \sigma(J)] = 1. \quad (4.8)$$

For every $a' \neq a$,

$$q_1[(I, a')] = 0,$$

so (4.7) implies

$$v[(I, a'), (J, b')] = 0 \quad \text{for every } b' \in A_J.$$

Symmetrically, all cells with $b' \neq b$ vanish. Applying the two conservation equalities gives

$$\begin{aligned} \sum_{a' \in A_I} \sum_{b' \in A_J} v[(I, a'), (J, b')] &= \sum_{b' \in A_J} v[\sigma(I), (J, b')] \\ &= v[\sigma(I), \sigma(J)] = 1. \end{aligned} \quad (4.9)$$

The only cell not already forced to zero is $v[(I, a), (J, b)]$, so that cell equals one. This proves (4.5). $\square$

Theorem 3 (Exactness). *Under the local triangle condition (3.2),*

$$\boxed{V = \Xi}. \quad (4.10)$$

Proof. Inclusion $\Xi \subseteq V$ was proved in (2.6). By Proposition 1, every vertex of V is binary. Lemma 2 shows that such a vertex is exactly the correlation plan generated by the deterministic profile (π_1, π_2) , and therefore belongs to Ξ . Since V is the convex hull of its vertices and Ξ is convex,

$$V \subseteq \Xi.$$

Combining the two inclusions proves (4.10). No timeability or chance assumption was used. $\square$

There is no circularity in this argument: the scaled-extension theorem is used to establish integrality, after which factorization and the equality $V = \Xi$ are proved directly from the conservation equations.

6 Exact polynomial-size equilibrium constraints

Let $u_i(z)$ be player i 's utility at terminal z . Let $\sigma_i(z)$ denote player i 's last sequence on the path to z , or $\emptyset$ if player i never acts. To avoid reversing coordinate order for player 2, define

$$v_i[\rho, \eta] = \begin{cases} v[\rho, \eta], & i = 1, \\ v[\eta, \rho], & i = 2. \end{cases} \quad (5.1)$$

Here ρ is always a sequence of player i , while η is a sequence of the other player.

For $I \in \mathcal{I}_i$, let $Z(I)$ be the terminal histories whose paths contain a node in I . For $a \in A_I$, let $Z(Ia) \subseteq Z(I)$ be the terminal histories taking action a at I .

6.1 Continuation realization plans

For a trigger τ , let $Y_{i,\tau}$ be the sequence-form polytope of continuation strategies available to player i after that trigger. It has a representation

$$Y_{i,\tau} = \{y \geq 0 : F_{i,\tau}y = f_{i,\tau}\}. \quad (5.2)$$

For an ex ante trigger, this is the ordinary sequence-form polytope rooted at $\emptyset$. For a trigger at I , it is the continuation sequence-form polytope rooted at I , with root weight one.

Explicitly, its equations set the root continuation coordinate to one and, at every continuation information set K , impose

$$\sum_{a \in A_K} y[(K, a)] = y[\sigma_\tau(K)], \quad (5.3)$$

where $\sigma_\tau(K)$ is the continuation predecessor relative to the trigger. Perfect recall makes this predecessor well-defined.

Every behavioral continuation strategy produces a point of (5.2) by taking products of its behavioral action probabilities along player i 's continuation sequences. Conversely, from a point y one recovers the behavioral probability

$$\Pr(a \mid K) = \frac{y[(K, a)]}{y[\sigma_\tau(K)]} \quad (5.4)$$

whenever the denominator is positive; arbitrary probabilities may be chosen when it is zero. Thus (5.2) represents exactly all continuation strategies relevant to payoff computation.

All coordinates lie in $[0, 1]$, so $Y_{i,\tau}$ is compact. It is nonempty because one may choose any action distribution at each continuation information set. Its number of variables and constraints is $O(N)$. For z below the trigger, write $y_\tau(z)$ for the realization weight of player i 's continuation sequence along z ; this is a coordinate, or a fixed linear selector of coordinates, in y .

6.2 Canceled trigger-payoff formulas

The trigger sets are

$$\begin{aligned} T_i^{\text{NFCCE}} &= \{\emptyset_i\}, \\ T_i^{\text{EFCCE}} &= \mathcal{I}_i, \\ T_i^{\text{EFCCE}} &= \{(I, a) : I \in \mathcal{I}_i, a \in A_I\}. \end{aligned} \quad (5.5)$$

For NFCCE, the payoff from an ex ante deviation $y \in Y_{i,\emptyset_i}$ is

$$D_{i,\emptyset_i}^{\text{NFCCE}}(v, y) = \sum_{z \in Z} p_c(z) u_i(z) v_i[\emptyset, \sigma_{-i}(z)] y_{\emptyset_i}(z). \quad (5.6)$$

The obedient payoff is

$$O_{i,\emptyset_i}^{\text{NFCCE}}(v) = \sum_{z \in Z} p_c(z) u_i(z) v_i[\sigma_i(z), \sigma_{-i}(z)]. \quad (5.7)$$

For an EFCCE trigger I , the player follows recommendations before I but elects to deviate before observing the recommendation at I . The triggered deviation and obedience contributions are

$$D_{i,I}^{\text{EFCCE}}(v, y) = \sum_{z \in Z(I)} p_c(z) u_i(z) v_i[\sigma(I), \sigma_{-i}(z)] y_I(z), \quad (5.8)$$

and

$$O_{i,I}^{\text{EFCCE}}(v) = \sum_{z \in Z(I)} p_c(z) u_i(z) v_i[\sigma_i(z), \sigma_{-i}(z)]. \quad (5.9)$$

For an EFCE trigger (I, a) , the player follows recommendations until receiving recommendation a at I , and then uses a continuation strategy rooted at I . After canceling histories on which the trigger does not occur, the exact inequality is

$$D_{i,(I,a)}^{\text{EFCE}}(v, y) = \sum_{z \in Z(I)} p_c(z) u_i(z) v_i[(I, a), \sigma_{-i}(z)] y_{(I,a)}(z), \quad (5.10)$$

against

$$O_{i,(I,a)}^{\text{EFCE}}(v) = \sum_{z \in Z(Ia)} p_c(z) u_i(z) v_i[\sigma_i(z), \sigma_{-i}(z)]. \quad (5.11)$$

All correlation coordinates in (5.6)–(5.11) are relevant. In particular, in (5.10) the actual deviating terminal may take an action other than a , but relevance depends on the underlying information set I , which lies on the terminal path.

We justify the cancellation and factorization carefully. The mediator samples a pure profile from the distribution represented by v . Before an NFCCE deviation, no part of player i 's sampled plan is observed or followed, so only the other player's sequence event remains; its probability is $v_i[\emptyset, \sigma_{-i}(z)]$. Before an EFCCE trigger at I , the deviator has followed precisely the own prefix $\sigma(I)$, giving the coordinate in (5.8). Before an EFCE trigger (I, a) , the player has followed the prefix and observed recommendation a , giving the coordinate in (5.10). Conditional continuation randomization is independent of the sampled profile and contributes the realization weight $y_\tau(z)$. Chance contributes $p_c(z)$.

In EFCCE and EFCE, all histories on which the specified trigger does not occur have identical probabilities and utilities under obedience and under the causal deviation. Subtracting the two full expected payoffs therefore cancels those histories and leaves exactly (5.8)–(5.11). In particular, an off-recommended-action EFCE terminal need not have a full probability described by a single correlation coordinate before cancellation: unchanged and triggered contributions may both be present. The unchanged contribution cancels, and the remaining triggered contribution is exactly (5.10). No division by the trigger probability is used, so zero-probability triggers cause no difficulty.

A pure causal deviation is determined by its trigger and its continuation plan. Randomized deviations are convex combinations of such continuation plans, and their payoffs are linear. Hence the standard trigger-deviation characterization gives the following exact conditions:

$$D_{i,\tau}^c(v, y) \leq O_{i,\tau}^c(v) \quad \forall i, \quad \forall \tau \in T_i^c, \quad \forall y \in Y_{i,\tau}. \quad (5.12)$$

The argument above proves the required version of that characterization directly for NFCCE, EFCCE, and EFCE.

6.3 Dualization

By collecting terminal terms, there are rational matrices $A_{i,\tau}$ and rational vectors $d_{i,\tau}$ such that

$$D_{i,\tau}^c(v, y) = (A_{i,\tau}v)^\top y, \quad O_{i,\tau}^c(v) = d_{i,\tau}^\top v. \quad (5.13)$$

These objects are obtained explicitly by iterating over the terminal histories in (5.6), (5.8), or (5.10), and over the corresponding obedient terminal set. Thus they have polynomial-size sparse descriptions.

For fixed v , the maximum deviation payoff is the linear program

$$\max \left\{ (A_{i,\tau}v)^\top y : F_{i,\tau}y = f_{i,\tau}, \quad y \geq 0 \right\}. \quad (5.14)$$

The exact dual is

$$\min \left\{ f_{i,\tau}^\top \lambda_{i,\tau} : F_{i,\tau}^\top \lambda_{i,\tau} \geq A_{i,\tau}v \right\}, \quad (5.15)$$

where $\lambda_{i,\tau}$ is unrestricted in sign. The primal feasible set is nonempty and compact, so the optimum is finite and attained. The standard strong-duality theorem for a feasible equality-form linear program with a finite optimum therefore applies, and the dual is feasible, attained, and has the same value. Consequently, (5.12) is equivalent to the existence of $\lambda_{i,\tau}$ satisfying

$$F_{i,\tau}^\top \lambda_{i,\tau} \geq A_{i,\tau}v, \quad f_{i,\tau}^\top \lambda_{i,\tau} \leq d_{i,\tau}^\top v. \quad (5.16)$$

There is one NFCCE trigger, $O(N)$ EFCCE triggers, and

$$\sum_{I \in \mathcal{I}_i} |A_I| = O(N)$$

EFCE triggers. Each block (5.16) has polynomial size.

For a rational terminal objective g , define

$$\ell^\top v = \sum_{z \in Z} p_c(z) g(z) v[\sigma_1(z), \sigma_2(z)]. \quad (5.17)$$

Under obedience, the event that both sampled plans prescribe the player actions on the path to z has probability $v[\sigma_1(z), \sigma_2(z)]$, so (5.17) is exactly the expected objective.

For any chosen concept c , an optimal equilibrium is therefore the solution of

$$\begin{aligned} & \max_{v, \lambda} \quad \ell^\top v \\ & \text{subject to} \quad v \in V, \\ & \quad F_{i,\tau}^\top \lambda_{i,\tau} \geq A_{i,\tau}v, \quad \forall i, \tau \in T_i^c, \\ & \quad f_{i,\tau}^\top \lambda_{i,\tau} \leq d_{i,\tau}^\top v, \quad \forall i, \tau \in T_i^c. \end{aligned} \quad (5.18)$$

By Theorem 3, the constraint $v \in V$ admits exactly the correlation plans of actual distributions over pure profiles. Hence (5.18) neither relaxes nor restricts the desired equilibrium set.

All sequence-form matrices have entries in $\{-1, 0, 1\}$. Every other coefficient is obtained from a polynomial number of products or sums of input chance probabilities, utilities, and objective values. A product along a terminal path has bit length at most the sum of the input bit lengths on that path, and adding polynomially many such rationals increases encoding length only polynomially. Thus (5.18) has polynomial dimension and polynomial bit complexity. Standard rational linear-programming algorithms solve it in time polynomial in $N + B$.

7 Compilation into a generalized correlation DAG

It remains to prove that the scaled-extension formulation is representable by a valid generalized correlation DAG, rather than merely by an unrelated linear program.

7.1 DAG-form decision problems

We use the following form of the DAG semantics from Zhang, Farina, and Sandholm [4, Appendix A]. A DAG-form decision problem is a finite directed acyclic bipartite graph with inactive coordinate nodes and active decision nodes. There is a root coordinate of weight one. Each nonroot coordinate node has one generating active predecessor. An active node may have several incoming coordinate nodes, and its outgoing arcs lead to the coordinates corresponding to its actions.

If the incoming coordinate weights at an active node t are $\{x_q : q \in P_t^{\text{in}}\}$, its active mass is

$$m_t = \sum_{q \in P_t^{\text{in}}} x_q, \quad (6.1)$$

and its outgoing coordinates satisfy

$$x_{t,a} \geq 0, \quad \sum_a x_{t,a} = m_t. \quad (6.2)$$

A one-action node is deterministic.

The relevant validity condition can be expressed as the following fork–merge path condition:

If two directed paths have a last common vertex, are internally vertex-disjoint thereafter, and later first meet again, then their last common vertex is active.

Thus two branches that later merge must have separated through different actions of a common decision, rather than having split after an inactive coordinate that could activate both branches simultaneously.

The external DAG scaled-extension theorem needed here is the following.

Theorem 4 (DAG scaled-extension semantics). *Let D be a finite alternating active/inactive acyclic graph with one root coordinate, one generating active predecessor for each nonroot inactive coordinate, and the fork–merge path condition above. Then its realization flow polytope is obtained, in any compatible topological order of its active nodes, by iterated scaled extensions. A node with k actions contributes a simplex Δ_k , scaled by the sum of its incoming coordinate weights; a one-action node contributes a singleton scaled by that sum.*

Appendix A of [4] proves this implication from a valid DAG to its scaled-extension semantics. It does not assert that every arbitrary scaled-extension chain automatically gives a valid DAG. We therefore verify the fork–merge condition explicitly below.

7.2 Construction from the scaled extensions

Consider the decomposition (3.3). Write each scale as

$$h_t(x) = \sum_{q \in P_t^{\text{in}}} x_q, \quad (6.3)$$

where a coordinate projection is represented by a singleton set P_t^{in} .

Create the root inactive node corresponding to the initial coordinate 1. For each block t , create an active node d_t , add an incoming arc from each previously created coordinate node in P_t^{in} , and create one new inactive child for every coordinate of X_t . If X_t is a simplex, impose

$$\sum_a x_{t,a} = \sum_{q \in P_t^{\text{in}}} x_q. \quad (6.4)$$

If $X_t = \{1\}$, create one child and impose

$$x_{t,1} = \sum_{q \in P_t^{\text{in}}} x_q. \quad (6.5)$$

All coordinates are constrained to be nonnegative.

Every parent of d_t was created before stage t , so the extension order is a topological order. The graph is therefore acyclic. Paths alternate between inactive and active nodes, and each inactive node has exactly one generating active predecessor.

Equations (6.4) and (6.5) reproduce the corresponding scaled extension exactly. If the right-hand side of (6.4) is positive, normalizing the new block by that value gives a point of the appropriate simplex. If it is zero, nonnegativity forces the entire new block to zero, as in the definition of a scaled extension. Thus, before invoking any DAG semantic theorem, direct induction already shows that the auxiliary flow equations project to $P_T = V$.

It remains to establish validity.

Definition 1. A *pure block assignment* chooses one vertex of every simplex factor in (3.3); singleton factors have their unique choice. It recursively sets the selected simplex coordinate equal to its scale and all other coordinates in that block to zero.

Lemma 3 (Binary pure assignments). *Every pure block assignment produces a 0-1 point of V . Consequently, whenever a scale has the form*

$$h_t(x) = \sum_{q \in P_t^{\text{in}}} x_q,$$

at most one coordinate x_q , $q \in P_t^{\text{in}}$, can equal one under a pure block assignment.

Proof. Beginning from the integer 1, every pure simplex choice either appends zeros and the integer scale $h_t(x)$, or a singleton appends that scale. Because every scale is an integer sum of earlier

coordinates, induction shows that all coordinates are integral. Every pure block assignment is a point of $P_T = V$, so (2.3) makes all its coordinates binary.

If two incoming coordinates of block t were one, then $h_t(x) \geq 2$. A pure simplex assignment would set its selected child equal to $h_t(x)$, and a singleton assignment would set its unique child equal to $h_t(x)$. In either case a coordinate of V would exceed one, contradicting (2.3). $\square$

Lemma 4 (Fork–merge path condition). *The DAG constructed from (3.3) satisfies the fork–merge path condition.*

Proof. Every inactive coordinate node is live: it can have weight one in some pure block assignment. Indeed, each such node represents a relevant coordinate $v[\sigma_1, \sigma_2]$. Choose a pure plan of player 1 realizing σ_1 and a pure plan of player 2 realizing σ_2 . Their deterministic correlation plan has that coordinate equal to one and belongs to V . Since this point is binary and $V \subseteq [0, 1]^R$, it cannot be a nontrivial convex combination of distinct points of V ; hence it is a vertex. Lemma 1, applied recursively, represents it by a pure block assignment. The root is live trivially.

Suppose for contradiction that the path condition fails. Then two paths last diverge after an inactive coordinate node c and first remerge later. Because every inactive node has a unique generating predecessor, the first remerge must occur at an active node d_t , through two distinct incoming coordinate nodes $q, q' \in P_t^{\text{in}}$. Between c and d_t , the active nodes on the two paths are disjoint.

Choose a pure block assignment activating c . On every active node of the first branch, select the outgoing child lying on the first path; do the same on the second branch. These choices are compatible because the active nodes on the two branches are disjoint. Inductively, each coordinate on both paths becomes one. At a branch active node, the incoming path coordinate is already one; Lemma 3 ensures that all other incoming coordinates are zero, so the scale is one and the selected child also becomes one. It follows that both q and q' equal one in the same pure block assignment.

This contradicts Lemma 3, because q and q' are both incoming parents of d_t . Hence no such inactive fork followed by a remerge exists, proving the path condition. $\square$

The graph now satisfies every hypothesis of Theorem 4: it is finite, alternating, and acyclic; every nonroot inactive node has one creator; and Lemma 4 establishes the path condition. Topological processing therefore reproduces precisely the scaled-extension chain (3.3). Its realization flow polytope projects exactly to

$$V = \Xi. \tag{6.6}$$

7.3 Size and construction time

There are $T = O(M)$ active nodes, and the total number of inactive coordinates is $O(M)$. A simplex block has at most $O(N)$ outgoing arcs. Each multi-coordinate scale in the Farina–Sandholm construction is a relevance-filtered marginal or descendant-backfill sum over at most $O(N)$ explicitly represented actions or sequences, so every active node has at most $O(N)$ incoming arcs. Therefore

$$|E(D)| = O(MN) = O(N^3), \quad |V(D)| = O(M) = O(N^2). \tag{6.7}$$

All DAG-flow coefficients are zero or one.

The relevance relation and the local triangle condition can be tested in polynomial time by constructing the parent-sequence bipartite graphs and checking that no edge has both endpoints

of degree at least two. Theorem 2 then constructs the scaled-extension order in $O(M)$ time once those graphs are available, and the DAG is obtained by writing the arcs in (6.3).

The equilibrium program (5.18) may use the DAG-flow constraints in place of the explicit condition $v \in V$. All coordinates appearing in its objective and incentive constraints are inactive DAG coordinates. If a particular correlation-DAG interface requires separate readout coordinates, they can be added by one-action copy nodes, which preserve the projection, the path condition, and polynomial size.

Proof of Theorem 1. Condition (TF) implies the local hypothesis of Theorem 2. The resulting scaled-extension decomposition yields $V = \Xi$ by Theorem 3. Lemmas 3 and 4, together with Theorem 4, compile this decomposition into a valid generalized correlation DAG whose size is bounded by (6.7). The exact trigger constraints and their dual formulations give the polynomial-size linear program (5.18) for each of NFCCE, EFCCE, and EFCE. Its dimensions and rational encoding length are polynomial in $N + B$, and its objective is the desired expected terminal objective by (5.17). Polynomial-time linear programming therefore proves all three claims. $\square$

8 Scope and conclusion

The construction never assigns global chronological times to information sets. Its topological order is the dependency order of the scaled-extension fill and backfill operations. Information sets may therefore occur at different tree depths, and non-timeable games are covered. Chance affects only the rational coefficients in terminal payoffs and not the structural decomposition, so arbitrary and zero-probability chance branches are also covered.

Games with no information sets give the one-coordinate polytope $\{v[\emptyset, \emptyset] = 1\}$. One-action information sets are represented by singleton blocks. Disconnected star components and isolated rank-zero information sets are included in the exact Farina–Sandholm construction. Thus none of these boundary cases requires a separate assumption.

The exponential examples for the old chronological correlation DAG show only that its unchanged prescription enumeration can be large. The present DAG instead factors prescriptions through star-guided simplex splits and deterministic descendant backfills. An extended algorithm can test the triangle condition, use this constructor when it succeeds, and otherwise fall back to the original general-game constructor.

Accordingly, the answer is affirmative: for every finite two-player perfect-recall triangle-free game, including non-timeable games with arbitrary chance structure, optimal NFCCE, EFCCE, and EFCE for any rational linear terminal objective admit a polynomial-size generalized correlation-DAG formulation and are computable in polynomial time. This completes the requested solution, with no remaining gap under the stated hypotheses.

References

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